

CAB is a major collaborative initiative led by the Department of Agriculture, Fisheries and Forestry and the Commonwealth Scientific and Industrial Research Organisation (CSIRO).

CAB was launched at the 2024 Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES) Outlook event.

CAB will convene Australian biosecurity experts to enable, investigate and support innovative biosecurity solutions to deliver a more robust, resilient, and digitally-enabled biosecurity system by 2030.

CAB is working with governments, industry and the community, to support the [DAFF Biosecurity 2030 roadmap](#) and achieve outcomes in the [National Biosecurity Strategy](#).

The goal is to improve long-term, national biosecurity outcomes by delivering innovative technologies, digital systems and capabilities that transform our system performances by 2030.

CAB will work with a broad range of stakeholders across the biosecurity system to deliver tools and technologies and become more responsive to new and emerging threats and system challenges.

Areas of focus

CAB work will span animal, plant, and environmental biosecurity work, including research and innovation from offshore to on-farm.

Our goal is to improve long-term national biosecurity outcomes by delivering innovative technologies, digital systems and capabilities that transform performance.

CAB will focus on:

- Detection and diagnostic technologies and platforms. Activities focused on understanding, detecting, and diagnosing priority exotic pests and diseases.
- Intelligence gathering platforms. Activities where biosecurity needs intelligence-based data to deal with threats and define better markets of trade.
- Intervention technologies. Any tool of technology needed to intervene or mitigate a real or potential biosecurity impact from on-farm to national level response, including sovereign vaccines, antimicrobials and associated biomanufacturing, novel biocontrol, and biopesticides.
- Decision support systems and modelling platforms. Activities supporting preparedness and emergency response, including supporting scenario planning, national response capability exercises and real-time outbreak responses.
- Social-economic support. Activities supporting effective and integrated human aspects of shared responsibility and effective governance across the system.
- Performance assurance mechanisms. Activities that enhance evaluation and monitoring (cost effectiveness and success analysis) and mechanism to test the performance of the system and assure progress is on target.

Collaboration

CAB is seeking foundation partners across the biosecurity sector. We are aiming to catalyse technology development, innovation, and adoption across Australia's complex biosecurity system. Collaborating with all Australian biosecurity stakeholders will be critical to achieve our goal. This includes state and territory governments, private sector and industry, non-government organisations, universities, research institutes and community members.

We will be holding information sessions for external stakeholders in the coming months.

Current projects

BeeRight

Varroa mite a danger to honeybees across the globe. In response to the recent outbreak, CAB is leading a partnership to detect varroa mites using BeeRight technology.

Members of this partnership include:

- Australian Capital Territory Government
- Queensland Department of Agriculture and Fisheries
- University of Canberra
- CSIRO
- Commonwealth Department of Agriculture, Fisheries and Forestry

BeeRight technology is an Internet of Things (IoT) solution that can be set up in any hive without damaging the hive or bothering the bees. More information on the project is available in our [BeeRight Press Release](#).

Lumpy Skin Disease

Australia is free from lumpy skin disease (LSD) and has never had an incursion of the disease.

CAB will invest significantly to prepare for emergency animal diseases such as lumpy skin disease. See more information regarding [lumpy skin disease](#).

CAB partners

Video: Catalysing Australia's Biosecurity

Speaker 1: Thanks to our strong biosecurity system, Australia is free from many of the world's most damaging pests and diseases. This system protects over \$50 billion dollars in tourism, our \$73.5 billion-dollar agricultural sector, 1.6 million Australian jobs across the supply chain, Australia's \$6.5 trillion dollars' worth of environmental assets and our way of life.

Speaker 2: In an increasingly connected world, the threats and impacts from pests, weeds and diseases are growing. It is estimated that invasive species currently cost Australia around \$24.5 billion dollars a year. Outbreaks abroad highlight the impact on agriculture, people movement and business activities. Certain biosecurity threats are already being felt in Australia. Future incursions could damage our way of life. The possibility of incursion is always present. We cannot be complacent.

Speaker 1: We need to evolve our biosecurity system to address the growing threats and impacts of pests, weeds, and diseases. CSIRO and the Department of Agriculture, Fisheries and Forestry are working together, with government, industry and the community, to support the Commonwealth Biosecurity 2030 roadmap and the National Biosecurity Strategy. Our goal is to improve long-term National biosecurity outcomes by delivering innovative technologies, digital systems and capabilities that transform performance by 2030.

Speaker 2: We will do this by working with a broad range of stakeholders across the biosecurity system to deliver tools and technologies that enable rapid pest and weed detection and identification, on-farm disease diagnoses and step change our capacity to prepare for and respond to incursions and reduce existing impacts.

Speakers 1 and 2: Together, we are Catalysing Australia's Biosecurity.

Related information

Australia has one of the strongest biosecurity systems in the world, free from many of the world's most damaging pests, weeds and diseases. But in an increasingly connected world, the threat to Australia from the impacts of these biosecurity risks continues to grow.

We need to evolve our biosecurity system to address these threats to protect over \$50 billion in tourism, our \$73.5 billion agricultural sector; 1.6 million Australian jobs across the supply chain and Australia's \$6.5 trillion worth of environmental assets and our way of life.

The impact of incursions will be significant. It is estimated that invasive species currently cost Australia around \$24.5 billion a year. This estimate does not include the ripple-effect of environmental impacts on jobs and our way of life.

For example, the direct economic impact of a large, multi-state outbreak of foot-and-mouth disease could reach \$80 billion over 10 years. While, if the exotic khapra beetle became established it could cost our economy \$15.5 billion over 20 years.

CAB will look for data-driven innovative solutions with partner agencies to address these threats.

CAB will support and enable innovative science and technology solutions to build a stronger biosecurity system as required by the [National Biosecurity Strategy](#).

The strategy identifies Australia's need to develop "a risk-based system underpinned by science that protects Australia's people, our environment and economy from the biosecurity threats of today and tomorrow".

It also highlights the important role of technology, research, and data in enabling this goal.

Together with government, industry, universities, community, and First Nation partners, we are looking to collaborate and invest in innovative science and technology outcomes.

By working together, we aim to deliver an integrated, secure, data-driven, and technology-enabled biosecurity system for Australia.

In its report [Value of Australia's biosecurity system \(unimelb.edu.au\)](https://unimelb.edu.au/research/centres-of-excellence/biosecurity-risk-analysis) the [Centre of Excellence in Biosecurity Risk Analysis](#) has estimated that a fit-for-purpose biosecurity system would provide \$314 billion worth of benefit to Australia over the next 50 years.

The [Australia's Biosecurity Future report](#) provides a transformational vision for a resilient biosecurity system in 2030, and outlines what needs to happen to get there. It focusses on the pressing need to transform rather than just scale up the current system.

Source: <https://www.agriculture.gov.au/biosecurity-trade/policy/catalysing-australias-biosecurity>